Microfracturing and the Inelastic Deformation of Rock in Compression

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The cracking that occurs during the deformation of rock in compression was studied by detecting and analyzing the radiated elastic waves. A new experimental method which increased the sensitivity of detection by several orders of magnitude over that of previous studies was used. The pattern of crack occurrence for a wide variety of rocks and at confining pressures up to 5 kb was found to be closely related to the stress-strain behavior. Dilatancy in the region above half the fracture stress was found to be directly proportional to cracking. Microfracturing, i.e. cracking, of brittle rock was compared with the microfracturing observed in frictional sliding and in the deformation of ductile rock. Cataclastic deformation of marble was found to be similar to frictional sliding in this respect but distinctly different than brittle deformation. The marble was found to undergo a gradual transition from cataclastic to fully plastic flow as confining pressure was increased in tests up to 4 kb. A model of deformation of an inhomogeneous brittle material is introduced which predicts the observed microfracturing activity and inelastic stress-strain behavior of rock. According to this model, microfracturing events in the dilatant region below about 95% of the fracture stress can be considered randomly independent. At higher stresses, where a rapid acceleration of activity is observed, the events cannot be considered independent and are correlated with the formation of the fault.

INTRODUCTION

For many geophysical applications the stress-strain behavior of rock at low temperature and pressure has been approximated by linear elasticity. It is well known, however, that at very low stresses the closure of pre-existing cracks within rock profoundly affects physical properties, such as elasticity [*Walsh and Brace*, 1966]. Although these effects usually disappear at a confining pressure of a few hundred bars, *Brace et al.* [1966] recently demonstrated that brittle rock is nonelastic at higher stresses, even at high confining pressure. This effect (Figure 1), which is noticeable at stresses greater than about half the compressive strength, is characterized by a decrease in modulus and by *dilatancy*. Here, dilatancy refers to an increase in volumetric strain relative to the increase that would be expected if the material were linearly elastic. The dilatancy was traced to the formation of small cracks within the rock.

This cracking was first detected indirectly by *Obert and Duval* [1942]. They found that small noises were emitted from rock during loading in compression, and they attributed these noises to cracking. These small cracking events, which we shall refer to as *microfractures* in order to distinguish them from fracture of the test specimen itself, seem generally to accompany the deformation of brittle rock. It would be interesting, however, to know if dilatancy can be accounted for solely by microfracturing or if other processes are involved. We would also like to know how this behavior agrees with our ideas of brittle fracture of rock and whether microfracturing also plays a significant role in frictional sliding or deformation of ductile rock.

If microfractures propagate in a brittle manner, we would expect that each event would radiate elastic waves of the sort detected by Obert and Duval and by later workers. The study of the elastic radiations associated with microfracturing offers an excellent means of observing directly the microscopic processes that occur during deformation. These processes can then be correlated with the observed macroscopic stress-strain properties. In this investigation, microfracturing was observed, and a pre-

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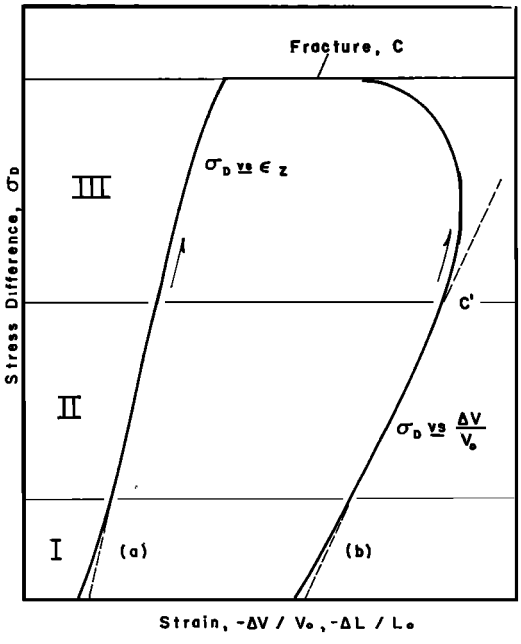


Fig. 1. Stages in the stress-axial-strain and stress-volumetric-strain behavior preceding fracture. Modified after *Brace et al.* [1966].

eise and systematic study of the influence of it on the macroscopic stress-strain behavior of brittle rock was made. These observations were then compared with the microfracturing observed during frictional sliding of rock and during deformation of a ductile rock.

Experimental Technique

Microfracturing detection. Previous investigators of microfracturing used the same basic technique. Events were detected with a piezoelectric transducer; the amplified signal was recorded with an audio-frequency tape recorder and displayed on an oscillograph or pen recorder. My investigations with a wide-band system [*Scholz*, 1967] have revealed, however, that most of the microfracturing radiation emitted is in the 100-khz to 1-Mhz range. Consequently, the limited frequency response of the systems used in previous work severely limited the observations.

The experimental system used in the present study is shown schematically in Figure 2. The essential component is a TMC model 102 pulse height analyzer. This analyzer has 100 channels and was operated for these experiments as a multiscaler. In this mode, each channel operates as a scaler for a predetermined time (dwell time). The analyzer thus counts the number of pulses above a certain threshold that occur during the dwell time, then switches to the next channel, and repeats the precedure until all channels are used. The number of events per channel is stored in a magnetic core memory and read out digitally after the experiment. A direct measurement of the number of events that occurred per unit time versus time is thus obtained. The dwell times used were 5 or 10

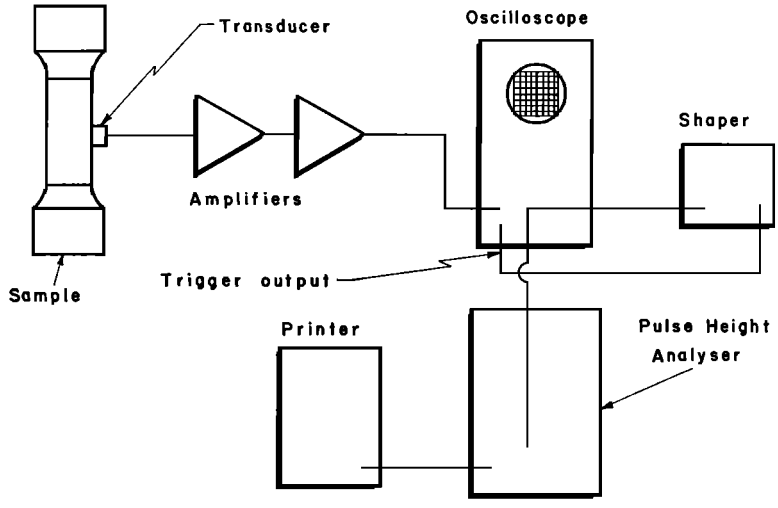


Fig. 2. Experimental system for the detection and analysis of microfracturing signals.

sec and were regulated by an external timing clock.

Preliminary investigations showed that typical microfracturing signals are in the form of decaying sinusoids with periods of a few microseconds and trainlengths of 10 to 100 μsec. Consequently, the signals could not be admitted directly to the analyzer, since more than one count would be actuated. Instead, the wave trains were shaped after amplification in such a way that the input to the analyzer was the envelope of the negative part of the wave train, which was found to give very satisfactory results. The design of the shaping circuit is described more fully in *Scholz* [1967].

Amplification was achieved in three stages. The first stage was a Hewlett-Packard model 465 amplifier, which has a 40-db gain and served as an impedance match, with 10-megohm input and 50-ohm output impedance. A HP 461 amplifier was used as the second stage. A variable gain third stage was obtained simply by using the trigger output from the monitoring oscilloscope, a dual trace Textronix 545 B, which was used to monitor both the signal after the second amplifier and the shaped signal. The shaper input was directed into a diode, so that the system noise in a particular experiment could be held below the diode cutoff by adjusting the third stage gain, and was noise-discriminated. The noise counted as events by the analyzer was in most cases reduced to a negligible level by means of this technique.

The transducers used in the experiments were barium titanate compressional mode disks, 6 mm in diameter and 2.5 mm thick. Copper foil electrodes were attached to the faces of the disks with conductive epoxy, and the package was cemented to a rubber pad and brass backing. Over-all frequency response of the amplification system was flat from 100 hz to 1 Mhz. This system was found to be extremely well suited to microfracturing studies. By simply increasing the frequency response, I found that the measurement of microfracturing activity with this system was two to three orders of magnitude more sensitive than the measurements of the previous workers. At the same time, the data processing time was negligible, so that a much more detailed and systematic study was possible.

Compression tests. Three types of sample designs were used in the experiments. Samples for uniaxial compression tests were of the form designed by *Mogi* [1966] to eliminate end effects. End effects seriously alter microfracturing behavior [*Scholz*, 1967]. These samples were right cylinders 1.6 cm in diameter by 5 cm long, attached to steel end caps by epoxy fillets. In the same work, Mogi found that end effects were negligible at confining pressures above a few hundred bars, so that specimens used for triaxial experiments were straight cylinders of the same size. Jacketing was 0.13-mm-thick copper.

Specimens for frictional sliding experiments have been described by *Byerlee* [1967]. Briefly, they were straight cylinders in which a flat ground sliding surface had been introduced at 45° to the axis. The two pieces were fitted together inside a 0.13-mm-thick copper jacket, and the ends were sealed with rubber tubing to steel end plugs. To allow the surface to undergo more than one stick-slip displacement without bursting the jacket, the copper jacket was strengthened around the margins of the sliding surface with a strip of RTV 102 silicone rubber sealant. The transducer was attached to a section of bare copper left exposed between this seal and the end seal. In all cases the transducers were clamped directly to the side of the specimens with rubber O-rings, with a film of Dow 276-V9 resin at the interface to ensure good coupling.

Volumetric strains were measured with strain gages described by *Brace et al.* [1966]. The loading machine has a ball screw loading mechanism that imparted a high stiffness (about 10^6 kg/cm). All experiments were done at a constant strain rate of 10^{-5} sec^{-1}, which was achieved by driving the ball screw with a constant speed motor.

Owing to the very high frequencies of the microfracturing signals and the design of the loading system, mechanical noise presented no problems during the experiments. Electrical noise was eliminated by shielding the sample, in uniaxial experiments, with 0.025-mm-thick copper. The pressure vessel itself served as a shield for tests under confining pressure.

Rocks studied. Six rocks were selected for the study. One, a medium-grained pure calcite marble, was described in the earlier study [*Brace et al.*, 1966]. This rock was ductile even

TABLE 1. Descriptions of Rocks

d is the average grain diameter. The total porosity η was determined by immersion in CCl_4.

Rock	Density, g/cm³	d, mm	η	Modal Analysis
Westerly granite	2.646	0.75	0.009	27.5 qu, 35.4 mi, 31.4 an_{17}, 4.9 mica
Rutland quartzite	2.643	0.30	0.005	91 qu, 7 or, 2 mi
Marble	2.688	0.20	0.013	99 ca
Colorado rhyolite tuff	1.77	0.01	0.41	20 qu, 20 gl, 60 an_{16}
Pottsville sandstone	2.541	0.20	0.030	46.4 qu, 41.0 or, 10.9 mica
San Marcos gabbro	2.813	1.00	0.002	69.7 an_{42}, 8.2 au, 7.4 ho, 11.7 mica

Abbreviations

qu, quartz; ca, calcite; mi, microcline; gl, glass; or, orthoclase; ho, hornblende; an, plagioclase with anorthite content; mica, muscovite, biotite, and chlorite; au, augite-hypersthene.

at room pressure. The other rocks were all brittle under the testing conditions. One of these, Colorado rhyolite tuff, had an extremely high porosity of 40%, mostly due to open vesicles. Of the other rocks, San Marcos gabbro and Rutland quartzite were very compact, whereas Westerly granite and Pottsville sandstone had appreciable crack porosity. Petrographic descriptions of the rocks are given in Table 1.

Observations

Uniaxial compression tests. Unconfined compression tests were made on six rocks to determine the effect of mineralogy and texture on microfracturing behavior. Figure 3 shows the frequency n of occurrence of microfracturing events versus axial strain for Westerly granite and Colorado rhyolite tuff. For comparison, the axial stress-strain curves are also shown.

The pattern of microfracturing for all rocks tested was very characteristic. The similarity in behavior for the two rocks shown in Figure 3, for example, is remarkable considering their strong disparity in physical properties, including strength. At low stress, a flurry of activity occurs; this activity soon dies down to a very low level. At stresses near approximately half the fracture strength, microfracturing activity begins to build up once more and steadily increases until just before fracture, when a very rapid acceleration of activity occurs.

This pattern correlates closely with the stress-strain behavior. At low stresses, in the region in which the first activity occurs, the stress-strain curve is nonlinear and concave upward. At higher stresses, where microfracturing activity is very low, the rock is nearly linearly elastic, and the resumption of activity corre-

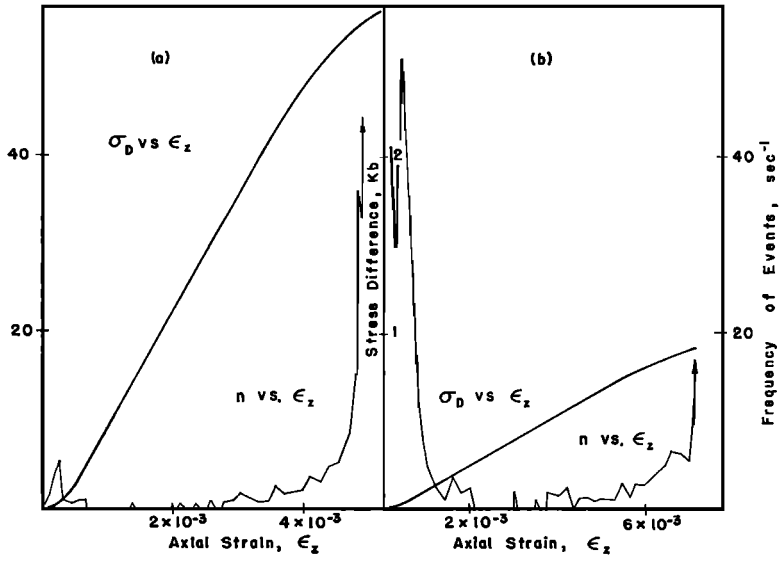


Fig. 3. Stress versus strain and microfracturing frequency versus strain for two rocks in uniaxial compression. (*a*) Westerly granite; (*b*) Colorado rhyolite tuff.

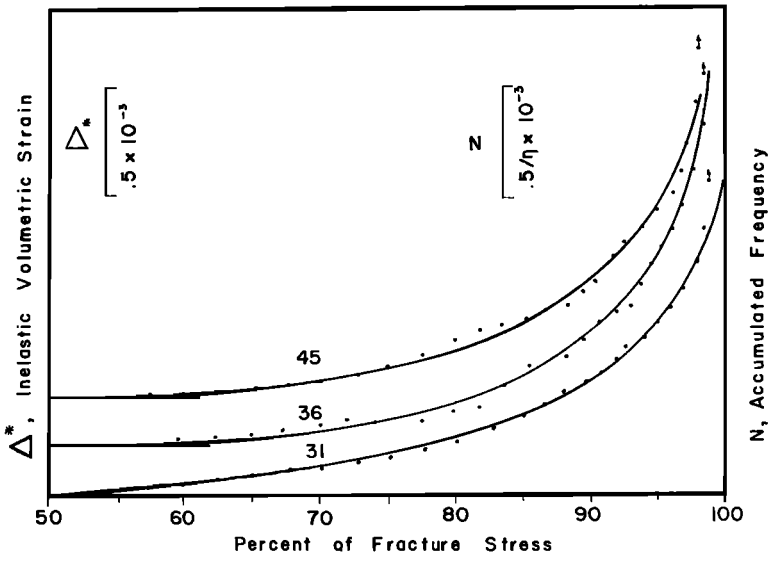


Fig. 4. Accumulated frequency (data points) of microfracturing events fitted to the inelastic volumetric strain (smooth curve) in the dilatant region for three uniaxial compression tests on Westerly granite. The values of η are given in Table 2.

sponds with the beginning of nonlinearity of the upper part of the stress-strain curve.

The only significant difference in behavior among the various rocks is in the amount of activity that occurs in the low-stress region relative to the activity occurring above half the fracture strength. It was found that the relative amount of activity detectable in the first region is roughly proportional to the porosity of the rock. Thus, in Figure 3, the amount of activity in the low-stress region of the compact Westerly granite is very slight relative to the activity that occurs later, whereas the activity in the first region of the highly porous Colorado rhyolite tuff reaches a very significant level. This observation is quite in accord with our knowledge of the behavior of rock in this part of the stress-strain curve. The inelasticity of rock at low stress is due primarily to the closing of pre-existing cracks and pores. (For a review of this behavior, see *Walsh and Brace* [1966].) Such adjustment of the void spaces would be expected to be accompanied by inelastic processes, such as crushing of pores and frictional sliding on crack faces, and is probably responsble for the microfracturing activity in this region. Consequently, a high degree of activity in this region would be expected from a high-porosity rock, as observed.

The nonlinear part of the stress-strain curve above about half the fracture strength is the region in which dilatancy occurs [*Brace et al.*, 1966]. Microfracturing was observed in this region, but we wished to test more quantitatively the hypothesis that dilatancy is due to the introduction of new void space produced by cracking. If it is assumed that the average size of microfracturing events remains constant throughout this region, the accumulated frequency of events N should be directly proportional to the inelastic part of the volumetric strain Δ^*, if this hypothesis holds. In this case, each event can be thought to contribute a small increment η to the volumetric strain. It is found [*Scholz*, 1968*a*] that the assumption of constant event size is not strictly true, but, for this application, the results will not be appreciably affected by its use.

In Figure 4 the results of three different uniaxial experiments on Westerly granite are shown. The ordinate is Δ^*, which is obtained by subtracting an elastic extrapolation from the measured volumetric strain as outlined by *Brace et al.* [1966]. N, the accumulated frequency of microfracturing events detected in the dilatant region, is best-fitted to these curves and is shown as data points. This was done by plotting N versus Δ^* over the entire range and

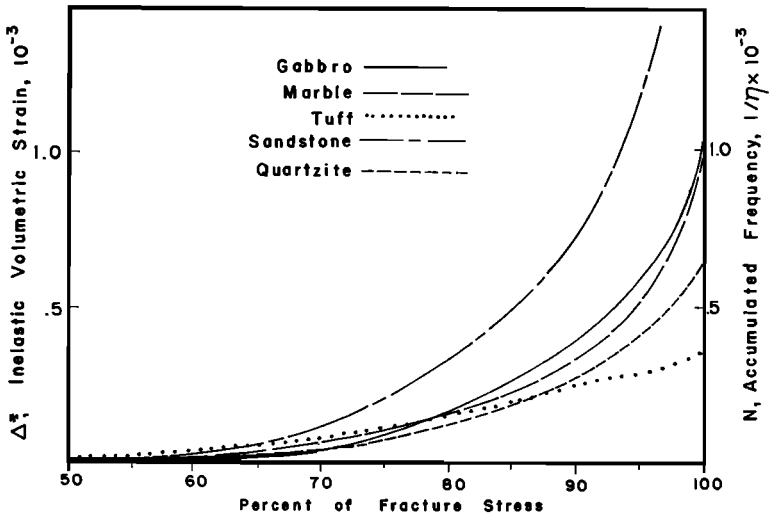


Fig. 5. Accumulated frequency and inelastic volumetric strain in the dilatant region for five rocks in uniaxial compression. Values of η are given in Table 2.

making a linear best fit. The correlation is very good, well within experimental error. A noticeable exception is the region above 95% of the fracture strength, where microfracturing activity rapidly accelerates. It is also interesting to note that the behavior is remarkably reproducible; the difference between these three experiments in all respects, including strength, is within experimental error.

The proportionality of N and Δ^* also holds for the other rocks tested. Although their strengths vary from 0.5 to over 5 kb, the behavior of all the rocks, when normalized with respect to fracture stress, is quite similar. The results for these rocks are shown in Figure 5. The data points have been omitted here for clarity. Variation of the proportionality constant η, given in Table 2, is considerable, owing to nonreproducibility of acoustical contact between experiments. The variation is particularly great for tests under confining pressure, when the gain was sometimes reduced to eliminate noise. Therefore, η is simply a constant of proportionality and has no quantitative physical significance. Also given in Table 2 is the quantity F, which is Δ^* when the stress equals 95% of

TABLE 2. Results of Compression Tests

Rock	Experiment No.	Pressure, kb	C, kb	C', kb	C'/C	F 10^{-3}	η 10^{-6}
Marble	30	0	0.50	0.28	0.57	0.52	1.70
Colorado rhyolite tuff	27	0	0.91	0.43	0.48	0.30	0.86
Pottsville sandstone	39	0	2.30	1.20	0.52	1.15	5.20
Rutland quartzite	38	0	5.05	2.94	0.58	0.40	4.15
San Marcos gabbro	43	0	2.15	1.16	0.54	0.58	*
Westerly granite	31	0	2.81	1.44	0.51	0.78	5.22
	36	0	2.82	1.50	0.53	0.88	6.69
	45	0	2.84	1.56	0.55	0.89	2.43
	61	1.00	7.90	4.34	0.55	1.66	13.30
	71	2.00	11.40	5.00	0.44	1.13	0.03
	63	3.00	13.60	5.16	0.38	1.96	2.64
	60	4.00	15.80	5.70	0.36	0.98	0.44
	62	5.00	16.80	7.40	0.44	1.20	2.46

* Excessive noise occurred during experiment.

the stress difference C at fracture. The quantity F is used as a measure of dilatancy because strains at stresses greater than $0.95C$ are likely to be nonuniform, and hence Δ^* would be nonreproducible and unreliable. C' refers to the stress difference at which dilatancy is observed to begin and C is the fracture stress difference. The uncertainty in C is a few per cent, 10 to 15% in C' and about 10% in F and η.

Triaxial experiments. Brace et al. [1966] concluded that confining pressure does not generally affect dilatancy. To see if the relation between microfracturing and dilatancy is affected by pressure, a series of experiments were conducted on Westerly granite at confining pressures up to 5kb.

A typical result is given in Figure 6. Aside from the noticeable lack of activity at low-stress differences, as we would expect from our above interpretation, the behavior is on the whole quite like that observed at atmospheric pressure. Comparisons of N versus Δ^* were made, and dilatancy was found to be related directly to microfracturing, as described above for the uniaxial experiments. The earlier conclusions that dilatancy is not appreciably affected by pressure are also confirmed.

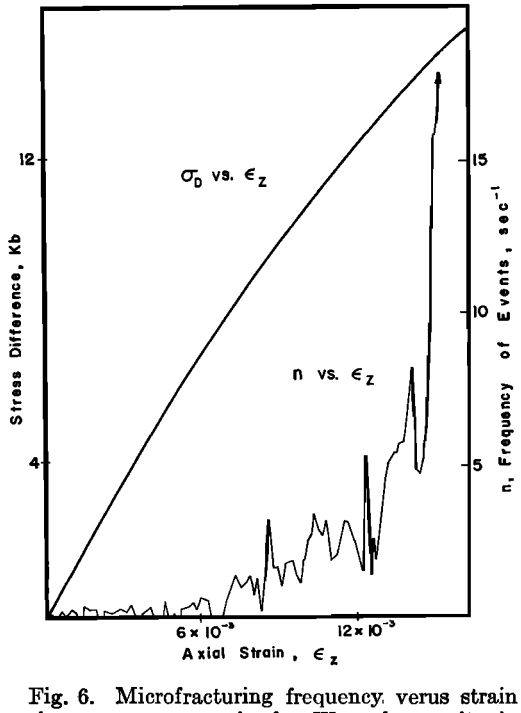


Fig. 6. Microfracturing frequency verus strain and stress versus strain for Westerly granite in triaxial compressions. Confining pressure, 4 kb.

Frictional sliding. Byerlee [1967] has recently suggested that friction in normally brittle materials is basically a brittle phenomenon. According to his model, frictional sliding takes place when the asperities on the sliding surfaces fail by brittle fracture. He suggested that two types of sliding take place: stick-slip, an inherently unstable form of sliding that is accompanied by a large stress drop, and stable sliding, which does not lead to detectable stress drops. The stable sliding often precedes the stick-slip. Because of the possible importance of friction in earthquakes [*Brace and Byerlee*, 1966], a study of the microfracturing that accompanies sliding was made to test this description of frictional sliding as a form of brittle fracture.

We show in Figure 7 the results of a typical experiment. The upper curve is a plot of force versus time. The solid parts of the curve are equivalent to force versus displacement, because strain rate is constant. The dotted vertical parts are stick-slip events. The curve, as a whole, is very similar to that described by Byerlee. Microfracturing, also given in Figure 7, shows a characteristic pattern during each stick-slip cycle. As force is increased, no events can at first be detected; apparently the surfaces are locked and no sliding is taking place. At some point prior to stick-slip, microfracturing abruptly begins to take place and continues at a roughly constant rate until stick-slip occurs. The arrows represent large numbers of events that were recorded during the stress drops. Immediately after the stick-slip, the level of activity falls again to a very low value, and on continued loading the cycle is repeated. The region of activity preceding stick-slip corresponds fairly closely with the slightly nonlinear parts of the force-displacement curves that Byerlee [1966] attributed to stable sliding. If this interpretation is correct, these results imply that the sliding begins rather abruptly and proceeds at a nearly constant velocity. There seems to be little indication that any acceleration of activity occurs before stick-slip, which is in marked contrast with the behavior during the deformation of intact rock described above. Intact rock is typified by a continual acceleration of activity during deformation.

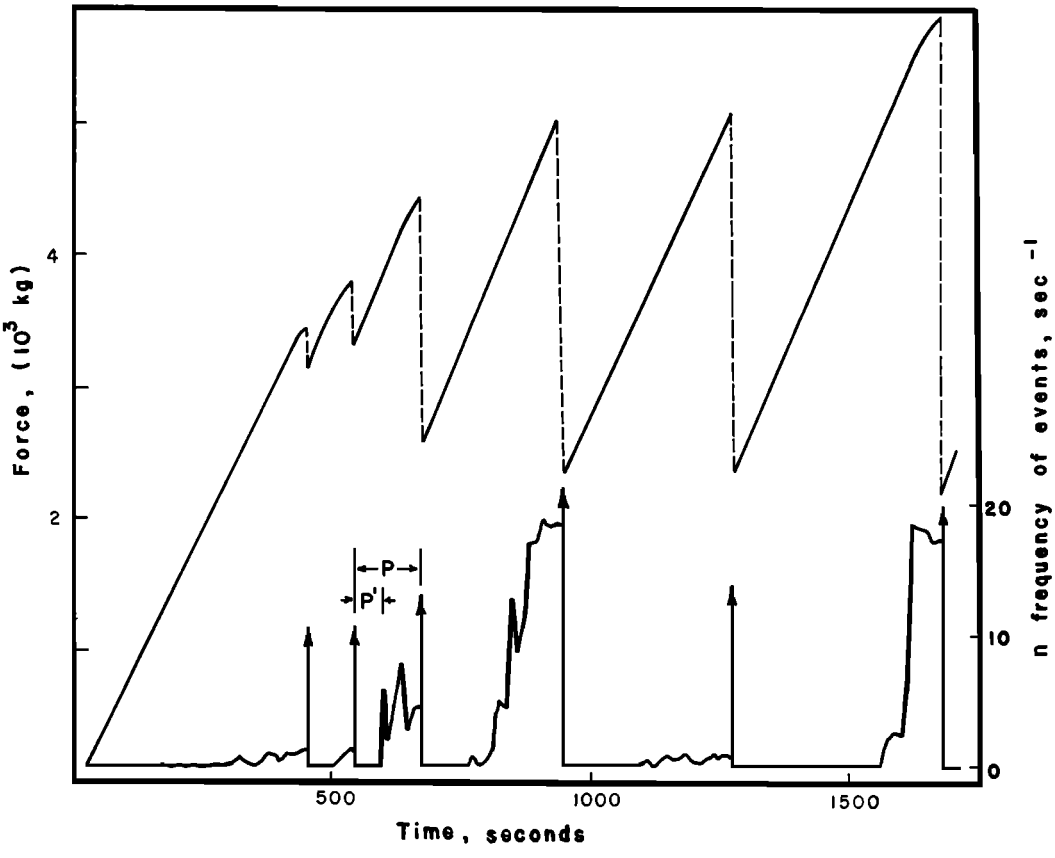


Fig. 7. Force versus time (upper curve) and microfracturing activity during frictional sliding of Westerly granite. Confining pressure 1 kb.

In Table 3 we list for the various experiments the mean frequency n of occurrence of events in the regions of activity preceding each stick-slip and the fraction of the stick-slip cycle P'/P at which this activity began. The behavior appears to be quite erratic. No pattern can be seen between successive stick-slip cycles during a given experiment or between different experiments. The level of activity may be high or low, may be continuous throughout the cycle, or may not occur at all. This lack of reproducibility was not observed in intact rock deformation and suggests that stable sliding is very structure sensitive; that is, it is affected by minor differences in the sliding surfaces.

Ductile deformation. It is generally assumed that at some temperature and pressure, most rock will become ductile; that is, a large amount of inelastic strain can be supported without fracture [*Griggs et al.*, 1960]. On the basis of microscopic examination, Griggs et al. have concluded that some plastic flow occurs during ductile deformation, but it has not been demonstrated that fully plastic flow, i.e. deformation produced entirely by dislocation motion, as in metals, is the mechanism of deformation.

The most detailed studies of rock ductility

TABLE 3. Results of Frictional Sliding Experiments

Experiment No.	Pressure, kb	1st Stick-Slip		2nd Stick-Slip		3rd Stick-Slip	
		n	P'/P	n	P'/P	n	P'/P
72	2.0	99	0.39	135	0.51	60	0.00
73	4.0	112	0.90	0	1.00	83	0.60
35	3.0	41	0.36	*			
33	1.0	†		56	0.50	4	0.20

* Jacket leaked after first stick-slip.
† First cycle interrupted by small stick-slips. (See Figure 8).

have been made on the carbonate rocks, such as Yule marble. These rocks are studied not because of their geologic importance but because carbonate rocks become ductile at relatively low temperature and pressure and hence are more readily amenable to experimental observation.

For the same reason, we have also chosen a carbonate for a detailed examination of ductility: the marble described above in the section on uniaxial compression tests. In so doing, we lose some generality in the applicablity of our results because silicate rocks will not behave identically, even at higher temperature and pressure, because of differences in the silicate and carbonate structures. In the light of the considerable experimental work that has been done on the carbonate rocks, however, it would be of interest to examine the role of microfracturing in the ductile deformation of these rocks.

It was shown above that the microfracturing of the marble in uniaxial compression was, just like the brittle rocks, directly related to the dilatancy. Marble, however, deforms cataclastically at room pressure, and the similarity of dilatancy and microfracturing to brittle rock, seen in Figure 5, is misleading. Plotting event frequency versus strain, as in Figure 8*a*, brings out the fundamental difference between the behavior of marble and the behavior of brittle rock. Comparing this with the stress-strain curve (Figure 8*a*), we see that microfracturing in the marble begins at about the start of the region of high curvature (yielding region) of the stress-strain curve and proceeds at a nearly constant rate with no acceleration, such as is observed in brittle rock (compare Figure 4). Similarly, the curves of N and Δ^* versus strain are seen to be linear. The distortion of the linear relation in the plot of Δ^* and N versus stress was simply due to the highly nonlinear stress-strain curve.

The eventual weakening of this rock is not accompanied by the formation of a fault but rather by a gradual loosening of the grains. After testing, the sample resembles a rather loosely packed sand. This type of deformation, termed *cataclasis*, is normally thought to be produced by stable intergranular frictional sliding. This view is supported by the rather strong similarity of the accompanying microfracturing to the microfracturing observed during stable frictional sliding. A steady rate of microfracturing is typical of both processes in contrast to the behavior of intact brittle rock. It was also found [*Scholz*, 1968*a*] that the frequency-magnitude relation of microfracturing events also sharply distinguishes frictional sliding and cataclastic deformation from that of brittle intact rock.

The marble was deformed in compression

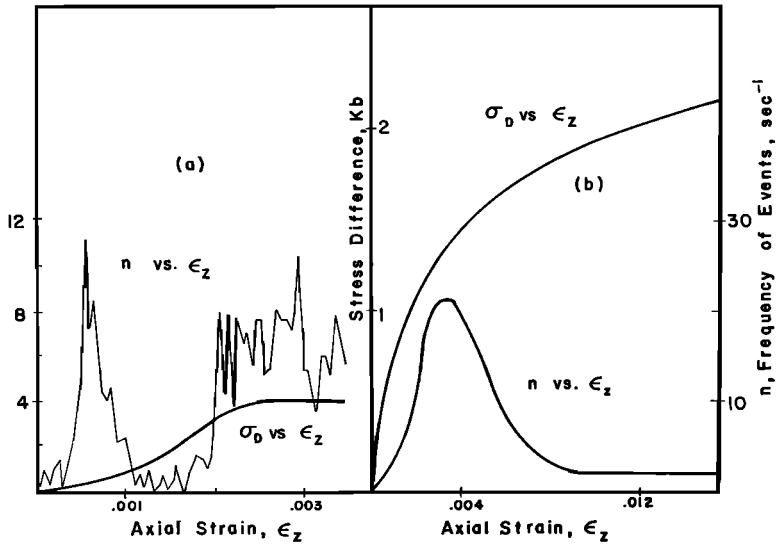


Fig. 8. Microfracturing frequency versus strain and stress versus strain for marble. (*a*) Uniaxial compression; (*b*) 1-kb confining pressure.

under confining pressures of from 0.25 to 4 kb. Under all pressures, the rock was found to be ductile in stress-strain behavior, and examination of the specimens after testing revealed no trace of a fault. Microfracturing activity under confining pressure was found to be quite different from that described above. The level of activity was quite low; therefore, the gain had to be increased to allow good detection. The result was that considerable noise was introduced. A count of background noise was made before each experiment with the specimen unloaded, and the final results were corrected for this ambient noise level. A smoothed curve of n versus strain and the stress-strain curve for a typical experiment under confining pressure are shown in Figure 8*b*. Activity begins at a very low-stress level, increases to a pronounced peak in the region of yielding, and decreases to a very low level in the nearly linear post-yield region. No definite change in this pattern was observed over the range of pressures at which tests were conducted, although the corrections for noise may have obscured small differences.

On the other hand, pressure had a pronounced effect on the stress-strain properties. The curves of the stress versus the axial strain for the experiments at each confining pressure are shown in the upper part of Figure 9. We see that pressure both strengthens the rock and increases the degree of strain hardening in the post-yield region. These effects are very rapid at first but diminish rapidly with pressure. Compare, for example, the change in the first kilobar of pressure with that in the second, or third. At 3 kb the pressure effect becomes negligible; at 3- and 4-kb confining pressure the stress versus axial strain curves are identical.

Coupled with this behavior is a very systematic change in the strain versus volumetric strain curves. These plots, in the lower half of Figure 9, are of *total* volumetric strain. The curves show the same general form, a small volume decrease occurs at low strains, the slope of the curve changes rapidly through the yield region, and finally the curve becomes linear after yielding. At low confining pressure, the large positive volumetric strain per unit axial strain reflects a large amount of dilatancy. The effect of pressure on the dilatancy is again quite similar to the effect of pressure on the stress-axial strain properties. The decrease in

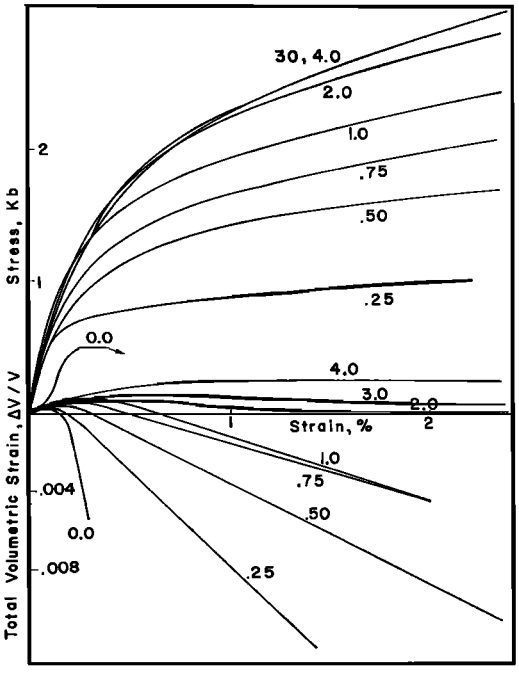


Fig. 9. Stress versus axial strain and volumetric strain versus axial strain for marble deformed under several confining pressures.

dilatancy resulting from an increase in pressure is large at low pressure and diminishes strikingly as the pressure is increased. At the highest pressure, 4 kb, there is no volumetric strain in the post-yield region.

Perhaps more diagnostic is Poisson's ratio, the ratio of lateral to axial strain, in the linear post-yield region. Poisson's ratio ν^*, the strength Y at 1% strain, the slope $d\sigma/d\epsilon$ of the stress versus axial strain curve after yielding, and the slope $d\Delta/d\epsilon$ of the volumetric strain versus axial strain curve in the same region are given in Table 4. These quantities are accurate to within 5%. Notice that ν^* decreases very rapidly as pressure is increased and appears to reach a limit of 0.5 at 4 kb. Almost all the change occurs in the first 2 kb.

Fully plastic flow of a polycrystalline material inherently implies some very general properties (see, eg., *Hill* [1950, chapter 2]). Specifically, plastic deformation is unaffected by confining pressure, provides little or no source for radiative elastic energy, and produces no volumetric strain. As shown above, all these conditions are very closely approached at confining

TABLE 4. Compressive Tests on Marble

Experiment No.	Pressure, kb	Y, kb	$d\sigma/d\epsilon$, kb	$d\Delta/d\epsilon$	ν^*
30	0	0.50	0.0	−4.70	2.85
65	0.25	0.87	11.9	−0.95	0.97
66	0.50	1.46	17.4	−0.51	0.75
76	0.75	1.76	28.8	−0.29	0.65
67	1.00	1.95	34.2	−0.33	0.66
70	2.00	2.24	43.5	−0.06	0.53
77	3.00	2.28	51.0	−0.06	0.53
68	4.00	2.28	50.5	0.00	0.50

pressures above 3 kb. The satisfaction of these criteria very strongly suggests that true fully plastic flow is the mode of deformation in this rock at high confining pressure. The mechanism of transition from cataclastic deformation to plastic flow is rather interesting. It is often said that the effect of pressure will be to inhibit cracking or frictional sliding, whereas the effect of temperature will be to enhance plastic flow. Therefore, an increase of pressure, temperature, or both will tend to produce plasticity. Just this effect of pressure is shown very graphically in the present experimental results. In the post-yield region, the linear relation between $\Delta v/v$ and ϵ_z shows that intergranular sliding produces a constant contribution to strain. As pressure is increased, intergranular sliding rapidly decreases and the material begins increasingly to resemble a fully plastic solid in that its stress-strain behavior is affected by pressure to an increasingly smaller degree. The close link between the decrease in dilatancy and the rate of change of mechanical properties with pressure can be seen from Figure 10, in which ν^*, the basic measure of dilatancy, is plotted versus pressure as are Y and $d\sigma/d\epsilon$. The change in ν^* with pressure is very smiliar to the change of Y and $d\sigma/d\epsilon$.

Some aspects of the microfracturing behavior need to be clarified with respect to this picture. Apparently some sort of reorganization takes place during the yield region that results in an equilibrium between plastic flow and cataclastic deformation. This reorganization is evidently accompaned by a brittle process, since a very distinct peak in activity is observed during yielding. Similar high rates of activity during yielding, followed by relative quiet, have been observed by *Shoemaker* [1961] in studies of the acoustic emission of steel and aluminum. His conclusions are that these emissions are due to plastic instabilities, such as the rapid release of dislocations from pile ups. These emissions cannot be entirely ruled out for marble, although at the same time brittle processes are not necessarily absent in metals. Twinning, in particular, may be a possible source of radiative energy in marble, although it should in general be accompanied by cracking.

The absence of microfracturing activity in the post-yield region, even when a large amount of dilatancy is observed, suggests that, if intergranular frictional sliding is the mechanism responsible for producing dilatancy, the plastic rather than the brittle form of friction obtains in grain to grain interfaces. This is in agreement with recent observations (J. D. Byerlee, personal communication, 1967) that many ductile rocks, including carbonate rocks, under moderate confining pressure exhibit a plastic form of friction that is not accompanied by stick-slip.

A number of workers have studied the brittle to ductile transition in rock. The most syste-

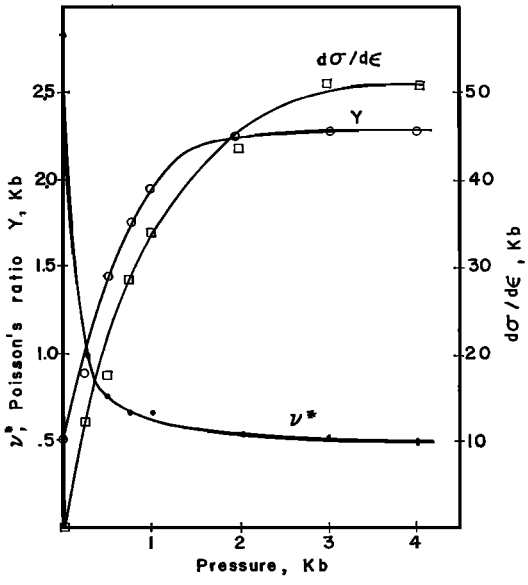


Fig. 10. The pressure dependence of the parameters Y, $d\sigma/d\epsilon$, and ν^* for marble. Y is the strength at 1% strain, $d\sigma/d\epsilon$ is the slope of the strain hardening curve, and ν^* is Poisson's ratio in the post-yield region.

matic study is that of *Heard* [1960], who defined the transition on the basis of the form of the stress versus axial strain curve and on the presence or absence of faulting evidence in the specimen after deformation. In this study we have seen that the transition is not sharply defined but that even the transition from cataclastic deformation to plastic flow for marble takes place over a quite large range in pressure (Figure 10). Also, the form of the stress versus axial strain curve and the appearance after testing yield little information as to the processes that actually occur.

Theory and Discussion

We have observed that microfracturing occurs in the deformation of brittle rock in compression and produces the characteristic inelastic properties of rock at high stresses in a way analogous to the way closure of pre-existing cracks produces inelasticity at low stress. This process of microfracturing is a very well defined property, quite similar for a wide variety of different brittle rock types. It is distinctly different from the processes that occur during frictional sliding or ductile deformation. These observations disagree with some of our previous ideas about deformation and fracture of brittle materials and serve as a clue to a better understanding of these processes. In this section, some of these differences will be explored, and a few very simple concepts developed to explain some of the properties of brittle rock.

Brittle fracture occurs by the separation of the material into parts. Yet any direct calculation of the forces necessary to break atomic bonds in a crystalline solid gives values for the strengths orders of magnitude greater than the strengths observed.

In his classic work, *Griffith* [1924] resolved this problem by assuming that defects in the form of narrow cracks producing stress concentrations exist in brittle materials. He proposed that fracture occurs when the stress at the tip of the crack with the highest stress concentration exceeds the stress necessary to break atomic bonds. This crack will then propagate unstably through the body, causing fracture. Using solutions for stress around an isolated crack in an elastic homogeneous body, he then calculated a failure criterion based on this model.

According to Griffith's model, a brittle material should behave elastically until the most highly stressed crack propagates and the sample fractures. This is clearly not what we observe for rocks in compression; cracks propagate at stresses much lower than the fracture stress, and many thousands of cracking events can be detected before fracture.

Brace and Bombolakis [1963] and *Hoek and Bieniawski* [1965] studied crack growth photoelastically in glass plates stressed in compression. They found that the initiation of crack growth under such conditions did not lead to failure but that the cracks became stable after propagating some fraction of their initial length. It might be suggested that these observations resolve the contradiction presented above: Griffiith's theory does not apply in compression because a crack in compression, unlike a crack in tension, cannot cause fracture by propagating unstably throughout the entire body.

If this argument is accepted, we would expect that microfracturing does not occur in tension, where Griffith's theory would apply. *Mogi* [1962] demonstrated, however, that in bending, which must involve tensile crack propagation, microfracturing was important. Indeed, the behavior he observed in bending is very similar in many respects to the behavior observed in the present study in compression. The microfracturing behavior of rock in uniaxial tension [*Brown*, 1965] is also very similar to the behavior observed in compression. The observed difference between the propagation of cracks in tension and compression therefore seems to have very little effect on microfracturing behavior. Cracks in tension as well as in compression do not propagate through rock as Griffith's model predicts, although it has been conclusively shown that tensile cracks behave this way in other brittle materials, such as glass. Why then, does rock behave differently than, say, glass?

The primary mechanical difference between these materials is that stress concentrations in rock do not arise solely from isolated cracks, as postulated by Griffith. Rock is a polycrystalline aggregate composed, in general, of several highly anisotropic phases. If a uniform stress is applied to such an inhomogeneous material, local stresses will fluctuate significantly from the applied value. The granular nature of rock complicates matters even further, since structural irregulari-

ties such as pores and cracks and interactions between them will produce additional fluctuations of the stress field within the body.

Consider how such an inhomogeneous body will react when subjected to a uniform increasing load. As the applied stress is raised, a point will be reached where, in the small region in which stress concentration is highest, the strength will be exceeded, and a crack will propagate. The crack will rapidly be arrested, however, if it propagates into adjacent regions in which the stress is lower. As the applied stress is increased still higher, the region with the next highest stress concentration will fail, and so on, until some sort of instability produces fracture of the entire body.

The formation of a macroscopic fracture in such an inhomogeneous medium, then, will be much more complex than the simple case given by Griffith. Since such a system is intractable by ordinary analytical techniques, we have chosen to describe the variation of stress within an inhomogeneous medium, such as rock, by a statistical model. Obviously such a model cannot describe the events that take place at a given point within the body but will be sufficient to predict the gross mechanical properties, which are usually of more interest.

To introduce the model, we take the simple case of an inhomogeneous brittle material under a single component of uniform applied stress. Such a system is illustrated schematically in Figure 11, where we show an inhomogeneous body under a uniform applied stress $\bar{\sigma}$. Now consider that regions within the body, small

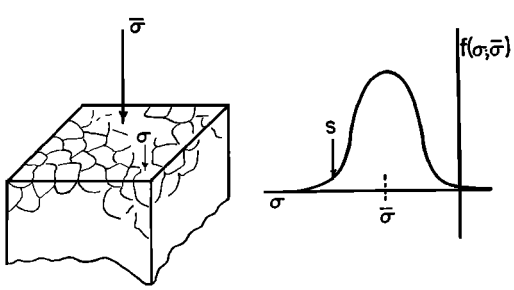


Fig. 11. An illustration of the response of an inhomogeneous body to an applied stress $\bar{\sigma}$ is shown on the left. The local stress σ may vary significantly from point to point. On the right we show an arbitrary stress probability function $f(\sigma; \bar{\sigma})$ and the local strength S. For explanation see the text.

enough so that the stress on them may be considered uniform, are subjected to a local stress σ, which varies in a random fashion throughout the body. Fracture will then occur within these small regions when σ exceeds S, the strength. It might be argued that this strength will also fluctuate, so that we must consider these variations as well. In fracture mechanics, however, it is not the stress or strength that is important but the difference between the two; therefore, either of them or both of them can be considered as variable. Since strength is conventionally thought of as a constant, it will be simpler to retain this concept, since the results will not be affected. The effect of cracks in producing stress concentrations will of course be much greater than the effects of other sources, so that it is likely that fracture will take place, in general, only in regions in which cracks exist. We may think of S, then, as defined by the presence of an average size and oriented crack. Variations of the size and orientation of cracks, as well as stress concentrations due to other sources, will be considered as variations of the local stress σ applied to such an average crack.

With such a model, σ can be considered to be a random variable. Therefore, the probability that the stress at a point is some value σ is given by a probability density function $f(\sigma; \bar{\sigma})$. Such an arbitrary function is shown in Figure 11. Also shown is our arbitrarily defined strength S. From equilibrium, the mean of $f(\sigma; \bar{\sigma})$ is $\bar{\sigma}$. Consequently, as $\bar{\sigma}$ is increased, $f(\sigma; \bar{\sigma})$ will move to the left. Eventually, there will be a finite probability that the stress in a region will exceed S, and microfracturing will begin. As $\bar{\sigma}$ is increased, the rate of microfacturing will steadily increase, since the frequency of microfracturing activity n during the increment $\bar{\sigma}$ to $\bar{\sigma} + d\bar{\sigma}$ will be given by

$$n(\bar{\sigma}) = K_1 f(S; \bar{\sigma})\, d\bar{\sigma} \qquad (1)$$

where K_1 is a constant. Accordingly, the accumulated frequency of events N is given by

$$N(\bar{\sigma}) = K_1 F(S; \bar{\sigma}) = \int_{-\infty}^{\bullet} K_1 f(\sigma, \bar{\sigma})\, d\sigma \qquad (2)$$

where $F(S; \bar{\sigma})$ is the probability distribution function of $\bar{\sigma}$. If we follow our previous assumption that such microfracturing events contribute an average increment η to the volumetric strain, then

$$\Delta^* = \eta K_1 F(S; \bar{\sigma}) \tag{3}$$

The statistical stress distribution, described by $f(\sigma; \bar{\sigma})$, is a fundamental property of a given rock and depends in detail on its composition and microstructure. It should also be pointed out that the mean microscopic strength S, although defined somewhat arbitrarily, also represents a distinct property of any given rock. Therefore, the gross microfracturing process is well defined by equations 1 to 3 and should be fairly reproducible, as observed.

It is interesting to take the analysis a bit farther by suggesting a possible stress distribution that might occur in rock. When microfracturing activity is fairly low, for example, the events may be considered to be isolated in space. The local stress which produces each event will then be an independent random variable, and $F(S; \bar{\sigma})$ can be approximated by a normal distribution.

The mean of the distribution is $\bar{\sigma}$. The stresses σ must vanish everywhere when $\bar{\sigma}$ vanishes (we disregard residual stresses that may be present, since they will be expected to be much smaller than applied stresses in the microfracturing region). A simple model satisfying this requirement is that the standard deviation μ is of the form

$$\mu = K_2 \bar{\sigma} \tag{4}$$

where K_2 is a constant. This relation implies that there exists a mean stress concentration factor ζ, where

$$\zeta = K_2 + 1 \tag{5}$$

In this model, the accumulated frequency of microfracturing is given by

$$N(\bar{\sigma}) = \frac{1}{2\pi K_2 \bar{\sigma}} \int_{-\infty}^{S} \exp\left[\frac{-(\sigma - \bar{\sigma})^2}{2(K_2\bar{\sigma})^2}\right] d\sigma \tag{6}$$

In Figure 12, we show this function fitted to the experimental data for uniaxial compression of Westerly granite. The fit is very good for stresses between 50 and 95% of the fracture strength, but the data cannot be fitted in the region above $0.95C$. This apparent change in

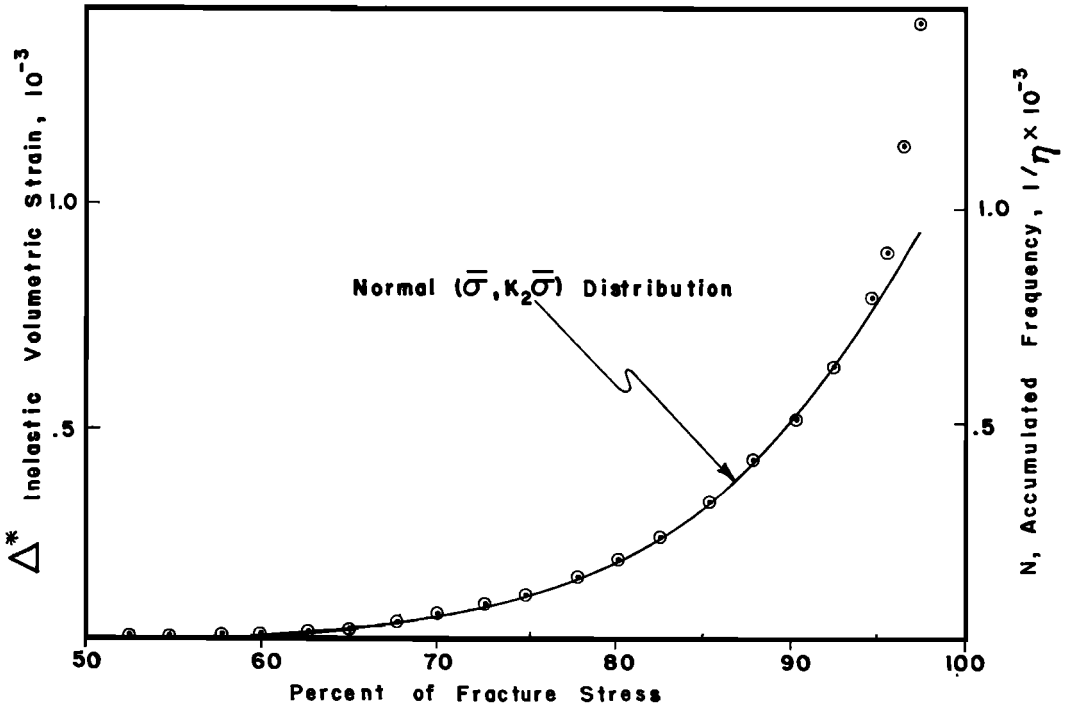


Fig. 12. The fit of a normal distribution with mean $\bar{\sigma}$ and standard deviation $k_2\bar{\sigma}$ to the accumulated frequency and inelastic volumetric strain observed in Westerly granite under uniaxial compression. Data points are values of Δ^*, to which N is proportional below $\sim 0.95\ C$. (See Figure 4.)

mechanism of the process at approximately $0.95C$ appears to be real. At this point rapid acceleration of microfracturing occurs. Above this stress it has been noted that the relationship between N and Δ^* does not in general hold and that these quantities are also not reproducible. In associated studies [*Scholz*, 1968*b*], we demonstrated, from more direct evidence, the difference between this region and the region at lower stress. By locating selected microfracturing events in space, I found that events at low stresses occur randomly throughout the specimen, whereas the events above $0.95C$ tend to cluster. These form the fault that leads to eventual fracture of the specimen. These observations agree with the results of the above statistical analysis, which suggests that events that occur in the region below this critical stress are randomly independent, whereas at greater stresses a larger number of events occur than are predicted by the theory, suggesting the formation of an instability. These two regions can be described as zones of *static* and *dynamic cracking*, respectively. From the best fit to the experimental data in the static cracking region, the parameters ζ and S for Westerly granite under uniaxial compression are found to be 2 and $2.25C$, respectively.

For simplicity, the analysis has been made only for the special case of a single uniform component of mean (applied) stress. It can readily be shown that the results will be the same for deformation in a uniform general mean stress system. In the general case, however, S will be a function of all components of mean stress, and all components of local stress will be random variables defined on a joint probability space. In the case of nonuniform stresses, i.e. where the mean stresses vary in space, the analysis will be somewhat more complicated, since the probability distribution of σ will be a function of position.

Conclusions

The technique developed in this study for detecting and analyzing microfracturing demonstrates that a great deal of information about the microscopic process accompanying deformation of rock can be obtained routinely. In particular, the conclusions of *Brace et al.* [1966] on the dilatancy of rock have been confirmed and expanded: for a very wide variety of brittle rocks of diverse structure and composition and over a wide range of confining pressure, deformation is accompanied by small-scale fracturing that begins at about half the fracture stress. As stress is increased, the cracking accelerates steadily until at a point a few per cent below the fracture strength a rapid increase occurs, marking the formation of the fault. The inelastic stress-strain behavior of these rocks can be completely accounted for by the strains contributed by this cracking activity.

These results may improve our understanding of the strength of brittle rock. We have shown that Griffith's criterion of brittle fracture is not applicable to rock because the inhomogeneity inherent in rock will produce fluctuations in the stress field sufficient to arrest cracks shortly after they have been initiated. Proceeding from these concepts, a simple statistical model was introduced, which predicts the observed microfracturing behavior. This theory does not, however, directly apply to rock fracture. As yet, the process that leads to the formation of the fault and to eventual fracture of the rock is not understood, nor are the factors that govern the stress at which this fracture occurs. Empirically, from the evidence of reproducibility of microfracturing, it is concluded that the fault begins forming when the crack density reaches some critical value. This is equivalent to the concepts of 'critical dilatancy' or 'critical void ratio' in the language of soil mechanics [*Scott*, 1963, p. 310]. The additional factor of the reproducibility of strength, coupled with the above observation, leads to the conclusion that, under identical conditions, the stress difference between fault initiation and fracture is constant.

These observations, of course, do not lead us very far toward an understanding of rock fracture, but they suggest an experimental approach. Our results have shown that for given conditions there is a 'critical crack density,' but it is not known how this depends on pressure and temperature, say, or time. The results reported for triaxial compression tests indicate that the pressure effect is slight but are inconclusive insofar as delineating the effect clearly. It is suggested that, if further work was done to determine the influence on critical crack density of these various parameters, an understanding of the process of fault formation and fracture might evolve.

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